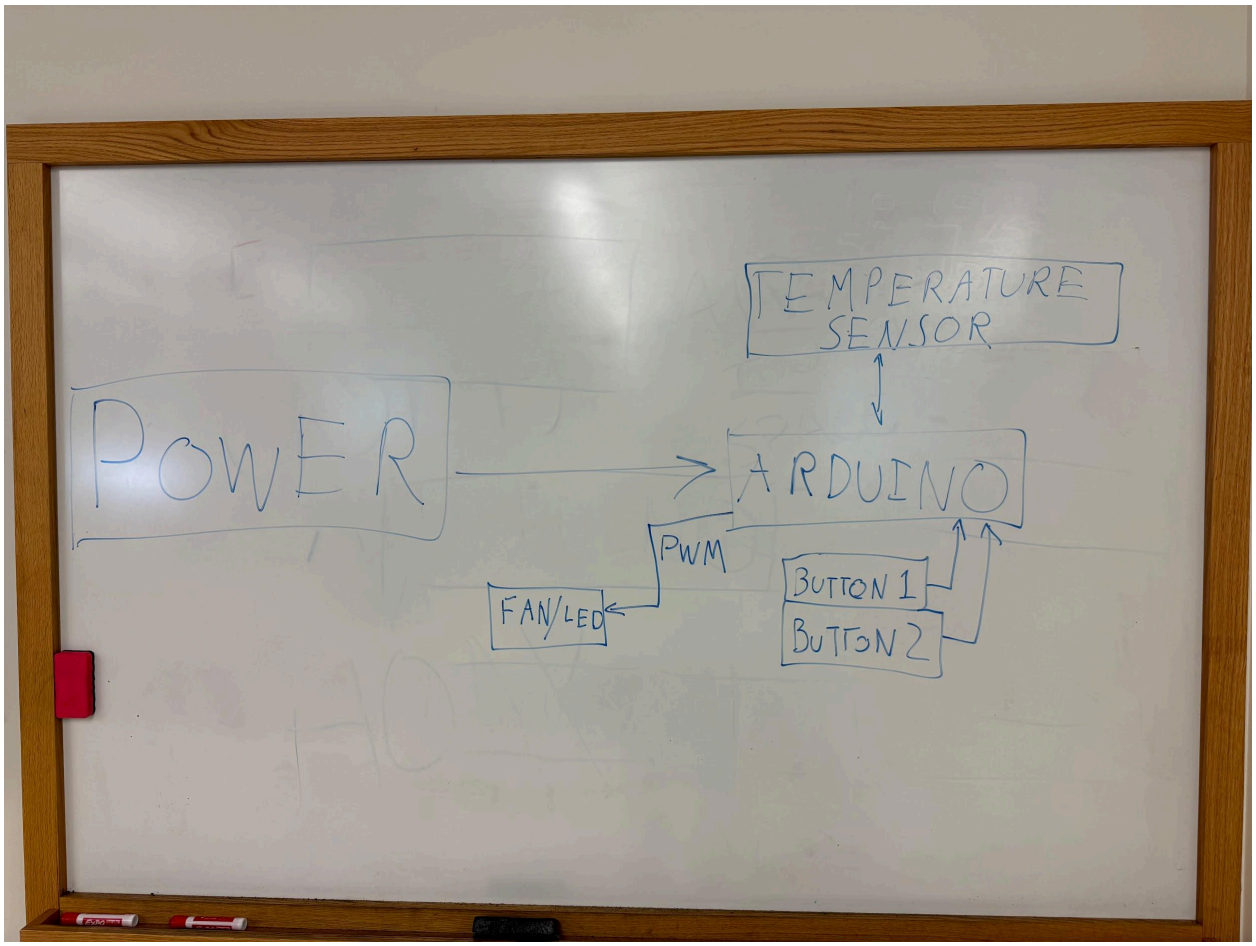This Arduino program implements a fan controller by using an analog temperature sensor, two pushbuttons, and a PWM output that drives an LED to simulate fan speed. The temperature sensor is connected to **pin A0**, where the Arduino's ADC collects **100 samples over a 5-second interval** and calculates an average value to provide a more stable and reliable temperature reading. This average ADC value is then converted into an estimated temperature, which is used to determine the required fan speed. The fan speed is simulated by changing the brightness of an LED connected to **PWM pin D9**, and the PWM frequency is configured to approximately **250 Hz** to meet the assignment requirement. **Button 1**, connected to **pin D2**, is used to turn the fan on or off, while **Button 2**, connected to **pin D3**, changes the fan sensitivity between three levels: low, medium, and high. The sensitivity setting changes how strongly the fan speed responds to temperature, with lower sensitivity producing a lower speed for the same measured temperature. Overall, this program demonstrates how an embedded controller can use ADC sampling, signal averaging, user input, and PWM output to implement a temperature-based fan control system.